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# Implementation of Neutral Wet Proppant Technology in Hydraulic Fracturing for Long Lasting Condensate Banking Solution in Deep Gas Wells

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## Abstract

Hydraulic fracturing is one of the solutions for the most gas production challenges from tight gas reservoirs which is condensate banking. Such a phenomenon usually results in sharp declines of gas and condensate production post fracturing and after reaching the dew point. This issue is more dominant in the depleted wet/condensate-gas fields, which making the post-fracturing cleanout complex and expensive due to the high capillary pressures slowing the completion brine cleanout. This paper summarizes the successful field application developed using neutral wet proppant in the hydraulic fracturing operation for deep tight sandstone gas wells to minimize the impact of condensate banking on hydrocarbon production in the Sultanate of Oman.

Thorough lab testing and screening processes to optimize the proppant selection and maximize the chances of success were conducted prior to taking the application to field implementation. In particular, the surface modifier material, proppant type and size were carefully selected to enhance the treatment. Also, to avoid any emulsion or incompatibility issue between the Neutral wet proppant and the formation fluid/rocks including condensate and highly saline formation water, a series of compatibility tests were conducted. In addition, careful quality and HSE planning was required before the field deployment due to cost associated with such proppant type.

A trial hydraulic fracturing campaign in 2 wells were conducted on one of the well-known fields that is impacted by condensate banking. The initial results of these trials show that, the implementation of the Neutral wet proppant technology managed to restore the gas production from closed-in well while increasing the condensate rate by more than 400% and gas rate by more than 410% from a new well compared to the expectations from the dynamic reservoir model of fractured well utilizing the offset wells frac model. These promising results have resulted in the planning of a wider deployment scope to further implement the Neutral wet proppant technology to optimize the production from several fields that are impacted by condensate-blockage impairment.

Using this Neutral wet proppant technology in the hydraulic fracturing operations as a long-term solution has showed improved efficiencies in time, production, and materials compared to the alternative periodical

remedial interventions. The application provides a more economical means to enhance production in tight gas fields of the Sultanate of Oman.

## Introduction

The global demand for energy continues to drive advancements in hydrocarbon extraction technologies. Hydraulic fracturing has become indispensable techniques for enhancing well productivity, particularly in tight gas reservoirs, and are widely applied in complex environments such as deep wet gas production operations. The success of these hydraulic prop fracturing stimulation treatments depends on creating and keeping highly conductive propped fractures, which allow oil and gas to flow from the reservoir into the well.

Despite continuous innovations and technologies in fracturing fluids and proppant materials, however the challenges remain. The primary concern is the significant retention of the fracturing fluids within the propped fracture and near wellbore (NWB) formation. This fluid retention is governed by capillary forces and the inherent wettability of the proppant and rock surfaces, which often exhibit a strong preference for water wet. When large volumes of viscous fracturing fluids are left behind due to the gelling agent insolubilities, which might build up in the proppant pack, leading to decreased effective permeability and reduced effective half length of the fracture. This phenomenon called water blocking, severely impairs gas flow and can significantly affect the production. The production from wet gas reservoirs that consist of gas and condensate is valuable for the economic point of view however, it has challenges in the production phase where the condensate banking challenge as the buildup of the condensate as a heavy material in the NWB region and it is more challenging situation with the depletion which lead to block the gas production and the well performance as overall.

Conventionally, surfactants designed as part of the frac fluid to reduce surface tension, offer transient wettability alteration and which mainly a liquid chemical that treat the proppant and flowed back with frac fluid and leaving portions of the proppant pack untreated. Other approaches post frac treatment, including solvent-based chemical treatments designed to alter wettability have shown promising results in restoring gas well productivity. These treatments enhance gas relative permeability by removing condensate banking and water blocking. A recent field application of solvent treatment in a gas well in Sultanate of Oman resulted in increased condensate and gas production rates. Altering the wettability of hydraulically fractured reservoirs has proven to be an effective approach in reducing condensate banking and water blocking. In hydraulically fractured wet gas wells, the likelihood of condensate banking forming in the NWB is high due to insufficient pressure support and lower bottom hole flowing pressure. There is a clear industry need for a more robust and permanent solution that can ensure efficient clean-up of all stimulation fluids and promote optimal dual phase of hydrocarbon flow for longer sustainable period.

This paper introduces the usage of a novel type of neutral wettability proppant, which developed to address the challenges of fluid retention and impaired hydrocarbon flow in hydraulic propped fractures due to both of condensate and liquid banking. This type of proppant is engineered to exhibit a "neutral" affinity for both condensate and water which becomes neither oil-wet nor water-wet. This unique surface property is achieved through advanced chemical coating modification, leading to a fundamental alteration of fluid and proppant interactions within the proppant pack to establish the potential of neutral wettability proppants for significantly enhancing post hydraulic fracturing stimulation fluid recovery, accelerating first gas production, improving long term productivity, and delivering substantial economic benefits of the gas well.

## Background and Theoretical Framework

Proppant Hydraulic fracturing is the process of injecting a high pressure frac fluid into the formation to create and propagate fractured channel where the proppant transported by the fracturing fluid during this process which stays within these fractures to keep them propped open after the injection pressure stopped. This creates highly conductive pathways that bypass NWB damage and easing hydrocarbon flow to the wellbore. The effectiveness of a propped fracture is quantified by its fracture conductivity (Cf), which is driven by the fractured proppant pack permeability (Kf) and the fracture width (Wf) which means the higher Cf leads optimal and sustained well productivity.

$$C_f = K_f \times W_f \qquad \text{(Equation 1)}$$

Wettability in the propped fracture describes the preference of a solid surface of the proppant pack and the formation to be in contact with one fluid rather than another when immiscible fluids are present which significantly influences the distribution and flow of fluids. The wet fluid will spread over it, while a non-wetting fluid will tend to bead up, minimizing contact. Figure 1 illustrates these behaviors, showing a wetting fluid spreading and a non-wetting fluid forming a droplet with a contact angle θ. This behavior is dictated by the interactive forces between the surface and the fluid as stronger adhesive forces lead to wetting, while stronger cohesive forces within the fluid lead to non-wetting.

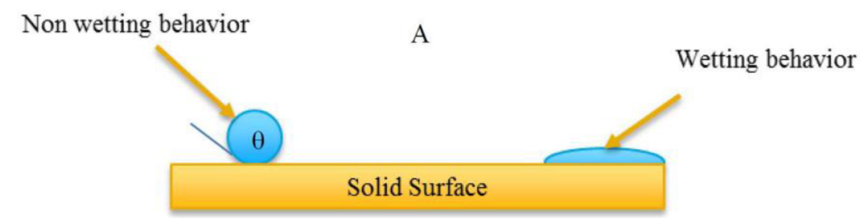


**Figure 1—Wetting behavior of surfaces**

The relationship between surface tensions at the solid liquid vapor interface is described by Young's equation (γLV cos θ = γSV – γSL), where γLV, γSV and γSL are the interfacial tensions between liquid to vapor, solid to vapor, and solid to liquid phases, respectively. The intrinsic contact angle θ is a key parameter. The surface roughness impacts the apparent contact angle (θ *). The Wenzel state in Figure 3A describes the fluid conforming to the surface topography, amplifying its intrinsic wettability. Conversely, in the Cassie state as in Figure 3B, the fluid is suspended on a mixed interface with air pockets, leading to superhydrophobicity.

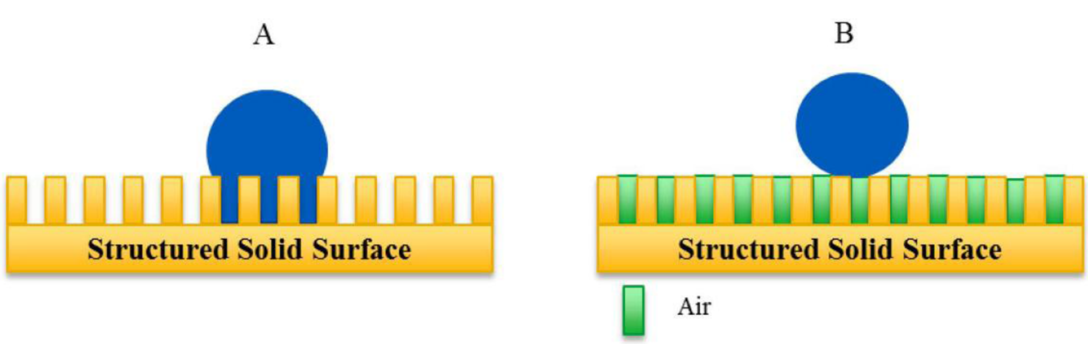


**Figure 2—Wetting on a structured surface (A) Wenzel state (B) Cassie state. In case B, air pockets are trapped into the textured surface.**

In a porous medium containing two immiscible fluids (e.g., condensate and water), capillary pressure (Cp) arises due to interfacial tension effects at curved fluid interfaces. Cp governs the distribution of these fluids and their ability to flow. The fundamental relationship for capillary pressure is given by Equation 2:

$$c_p = \mathrm{A} \cdot \frac{2\gamma\cos\theta}{a} \qquad \text{(Equation 2)}$$

Where Cp is the capillary pressure (psi), γ is the interfacial tension (dynes/cm), θ is the contact angle (degrees), a is the pore radius (microns), and A is a constant ($145.10^{-3}$) that converts units. Crucially, Equation 2 demonstrates that if the contact angle θ is 90°, the capillary pressure becomes null. In such a neutral wettability state, neither fluid is preferentially retained by the pore surfaces. As shown in Figure 3, the blockage of pore space by retained fluids due to capillarity can severely impede hydrocarbon migration, while the suppression of capillarity allows fluids to migrate easily. When surfaces are modified to be neutral, the intermolecular interactions between the solid and the fluids are minimized, facilitating easier extraction of fluids and leading to a lower differential pressure required for fluid movement which helps in the depleted gas wells that has a very high CGR ratio.

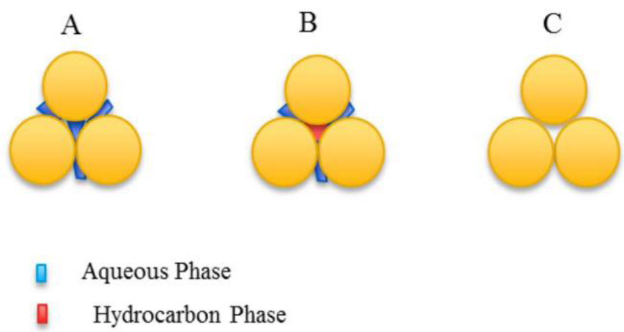


**Figure 3—Capillarity effect on flow of hydrocarbons. Blockage of pore space due to inadequate clean up of the fracturing fluid in the proppant pack or water blocks will lead to decrease of hydrocarbon flow, while if the capillarity is suppressed, the fluids can migrate easily.**

Historically, various methods have been employed to improve proppant pack conductivity and well cleanup:

- Surfactants: While effective at reducing surface tension and maximizing load water recovery (Phillips et al., 1984), surfactants provide only temporary wettability alteration (Weaver et al., 2010). They tend to plate off quickly, leaving portions of the proppant pack inadequately treated, and the temporary wettability changes can still lead to water blocking.
- Resin Coatings: Resin-coated proppants have shown success in increasing conductivity by reducing proppant flowback and mitigating polymer damage (Weaver et al., 2005). Some also help encapsulate fines. However, they are not primarily designed to optimize the dual-phase flow of oil and water and may not fully address capillary trapping.
- Superhydrophobic Coatings: These coatings, whether applied directly or as additives, are beneficial in immobilizing fines and improving cleanup time (Raysoni, 2012; Weaver et al., 2010). However, if the surface becomes highly hydrophobic (water-repelling), it often becomes oil-wet, potentially hindering oil flow and affecting overall permeability and conductivity. A mechanism that optimally decreases capillary forces for both oil and water is essential to prevent either phase from being trapped.

## Neutral Wettability Proppant Technology

The neutral wettability proppant employed in this study is based on High-Strength Ceramic Proppant (HSP) with a 20/40 mesh size, selected for its high mechanical strength, uniform sphericity, and consistent performance under downhole conditions. This proppant serves as the typical material for hydraulic fracturing operations in the analog wells. The surface modification process is engineered to permanently enhance the proppant's wetting characteristics while preserving its essential physical and mechanical properties, which were tested in the lab as below:

**Table 1—Physical and mechanical properties of HSP 20/40**

| Sieve Distribution | |
|---|---|
| US Mesh | Wt% Retained |
| 16 | 0 |
| 20 | 3.4 |
| 25 | 51.4 |
| 32 | 42.3 |
| 35 | 2.9 |
| 40 | 0 |
| 50 | 0 |
| Pan | 0 |
| Total | 100 |

| Physical Properties | |
|---|---|
| Apparent Density (g/cc) | 3.64 |
| Bulk Density (g/cc) | 2.1 |

| Crush Resistance | |
|---|---|
| API CRUSH @ 15K psi (% BWOW) | 4.3 |

The modification on the proppant lies in its permanent surface modification. This is achieved through the covalent bonding of specially engineered molecules to the oxide surfaces of the HSP proppant. These molecules are uniquely characterized by simultaneously possessing hydrophobic (water-repelling) and oleophobic (oil-repelling) properties, resulting in an amphiphobic surface which ensures that the proppant has minimal affinity for either water or hydrocarbons which translates to minimized intermolecular interactions between the proppant surface and flowing fluids. The neutral wettability proppant offers several advantages over conventional proppant technologies:

- **Flow Optimization**: the neutral proppant wettability ensures that neither phase adsorbed or trapped within the frac packs. This facilitates the simultaneous flow of both condensate and water from wet gas well, maximizing the overall permeability and conductivity of the proppant pack to both phases which is crucial for optimal production and increase the recovery.
- **Enhanced Fines Control**: Beyond reducing the frac fluid gel damage and condensate blocking, the surface modification can also impact the generation and migration of fines. As noted in laboratory tests, the amount of fines generated after crushing can be reduced, and turbidity can significantly decrease. This reduction in fines helps maintain long-term fracture conductivity and prevents pore throat blockage.

## Screening and Selection

Prior to field application, screening and selection process was implemented to identify the optimal neutral wettability proppant for deployment to be able to get all the mentioned advantages of this proppant. Samples from two different vendors of neutral wettability proppant were acquired and subjected to comprehensive laboratory testing with the local testing apparatus. This comparative analysis involved evaluating key performance indicators such as wettability to water and oil, permeability and flowback simulation and fluid compatibility.

## Experimental Methods and Materials

All proppant samples were split according to ISO 13503-2 recommendations prior to testing to ensure representative samples and the material used for the test are as below:

- **Proppants:** High Strength Ceramic Proppant (HSP), 20/40 mesh and Neutral wettability proppant from different vendors for better comparison.

- **Fracturing Fluids:** Broken crosslinked fluids, representative of what we use in the hydraulic fracturing operations, were prepared for standard compatibility tests.
- **Test Fluids:** Treated water, mineral oil and field condensate sample were utilized for laboratory tests.

## Wettability Determination

A subjective evaluation for amphiophobic behavior (exhibits both hydrophobic and oleophobic characteristics) on proppant surfaces (ie, a surface that repels both water and oil) were conducted on all samples. An untreated sand or ceramic proppant should absorb water and oil immediately upon contact. A proppant treated with a neutral wettability coating should repel both water and oil.

The followed procedure considers dropping the tested fluid onto a flatten surface (so that the droplets do not roll off) of the proppant sample. The surface must not be shaken or vibrated and observe the droplet for 5 minutes. If either droplet is absorbed within 5 minutes, the proppant fails, if droplet remain on the surface, either slightly elongated, or spherical, perhaps having an acute contact angle, the proppant passes. Tested fluids were water, mineral oil and condensate.

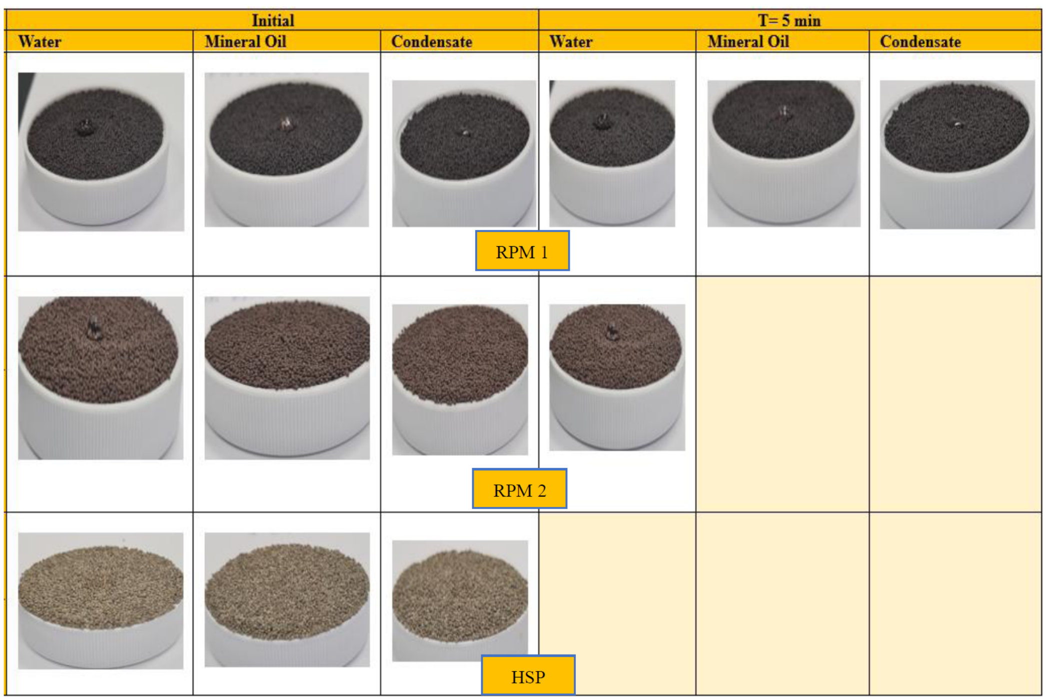


**Figure 4—Wettability of amphiophobic behavior of proppant vs. water, mineral oil, and condensate**

## Permeability and Flowback simulation

An in-house arrangement was conducted in the lab to simulate and quantify fluid displacement through the neutral wettability proppants. The purpose was to mimic the potential behavior of the fluids during the flowback and production operations. Two burettes were packed with 80 mL of sample the tested proppants to simulate a vertical fracture. One burette was filled with non-treated 20/40 proppant to be used as blank.

Figure 5—Column flow testing for different type of neutral wet proppant, that we flowed Water, crude oil through them.

The proppant pack in the burettes was filled with treated water and allow soaking for 30 minutes before allowing it to be flown by gravity. The pack was then displaced with mineral oil. The recovered volume and filtration time of each fluid (water and mineral oil) were recorded. The calculated flow rates (mL/min) and recovered volume (%) were calculated.

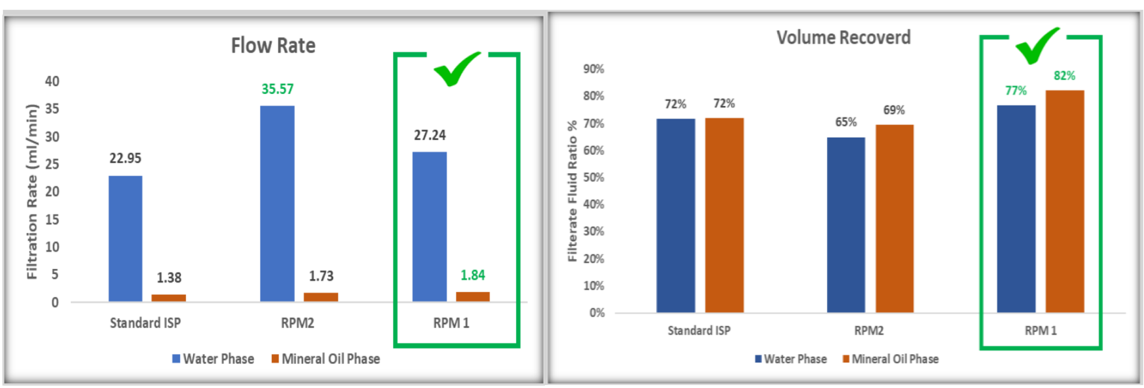


Figure 6—Column flow testing for different type of neutral wet proppant, that we flowed Water, crude oil through them.

As per these initial lab tests, each sample showed varying results as showed above in figure 4 and 6. The neutral wettability proppant RPM-1 demonstrated better performance across all critical parameters, particularly in achieving consistent neutral wettability and higher fluid recovery rates. RPM-2 showed a better result in the flowrate test, however considering the total volume recovers, RPM-1 still showing a better performance. This product was selected for the field pilot due to its significantly more favorable results. This screening ensured that the most effective proppant technology was deployed to maximize the potential benefits in the field.

## Laboratory Compatibility Testing

To determine whether RPM-1 and condensate are chemically compatible with the commonly used broken frac fluid (guar-borate type) under simulated downhole conditions, compatibility tests were conducted. Samples were aged at 50°C for 24 hours, followed by visual monitoring and filtration through a 100-mesh screen to detect any emulsion, sludge, or precipitate formation. All samples exhibited clear phase separation with no signs of emulsion, sludge, or solid precipitation, confirming the compatibility of RPM-1 and broken frac fluid and indicating that condensate addition did not impact system stability.

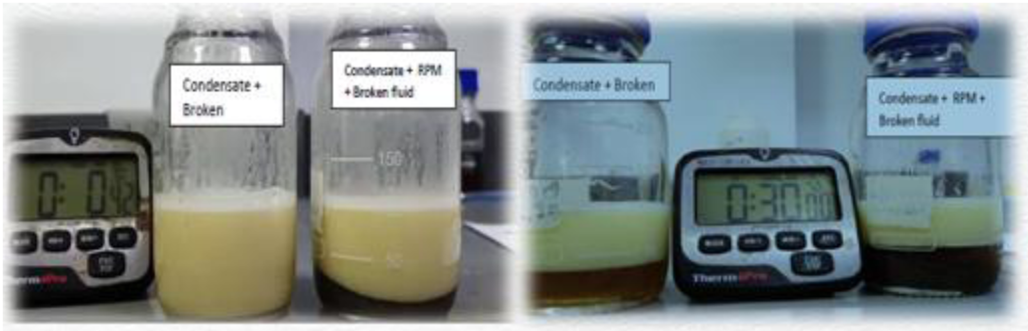


**Figure 7—Compatibility Test at 23 C**

**Table 2—Compatibility test at 23 C**

| | % Emulsion Breakout at 23 deg C (Minutes) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 1 | 5 | 7 | 10 | 15 | 20 | 30 | 90 |
| Condensate Sample+ Broken Frac Fluid | 20% | 40% | 50% | 60% | 65% | 70% | 80% | 93% |
| Condensate+ Broken Frac Fluid + **RPM-1** | 40% | 70% | 80% | 95% | 96% | 97% | 100% | |

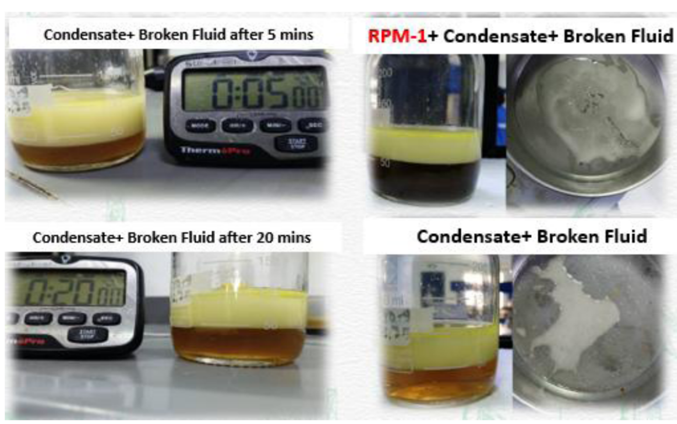


**Figure 8—Compatibility test at 50 C**

**Table 3—Compatibility test at 50 C**

| | % Emulsion Breakout at 50 deg C (Minutes) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 1 | 5 | 7 | 10 | 15 | 20 | 30 | 90 |
| Condensate Sample+ Broken Frac Fluid | 20% | 50% | 70% | 90% | - | - | 100% | |
| Condensate+ Broken Frac Fluid + **RPM-1** | 80% | 100% | | | | | | |

## Field Application and Data Collection

Field applications of the neutral wettability 20/40 HSP proppant were conducted cemented liner completions in the Sultanate of Oman, specifically targeting AMN formations which is part of the Haima Super Group in the FKR field which is located in the Ghaba salt basin and were formed as a result of Paleozoic salt diapirism.

The well completions involved standard procedures, with no modifications to equipment or processes typically used for conventional proppants. This ensured a direct comparison of proppant performance. Flowback data (cumulative water and condensate recovery, flow rates) and long-term production data (daily condensate and gas rates, water cut, flowing tubing pressure) were collected for the wells utilizing the neutral wettability proppant and compared against offset wells in the same field that used similar but untreated HSP proppant.

AMN Formation is Cambrian in age and forms part of the Haima Group. The current interpretation is that the AMN Formation comprises a mixture of fluvial, sabkha and aeolian sediments deposited within a low relief extradunal sabkha sandsheet to margin setting. Despite the misconception that the reservoir is of thick sand body with one hydraulic unit; stratigraphically, it is divided into different geological sandstone sub-units S-30, S-20 and S-10 which are inherited geological heterogeneity in addition to mechanical variation between these subunits, it has been proven by difficulty to hydraulic propped fracturing uniformly. This difficulty came from the. This has impacted the success rates in terms of placement, production and frac height, between wells with similar porosity and lithology.

All the wells in this field were completed as vertical cemented completion targeting AMN formation with 4-1/2" tubing, with depths ranging from 4300-4600 m MD. Two Proppant Hydraulic Fracturing stages were executed in this well targeting the same formation at different depth employed zirconate crosslinked frac fluid system with KCl substitute that is equivalent to 4% KCl base frac fluid. Proppant concentrations ranged up to 6ppa in the 1$^{st}$ stage and 5ppa in the second stage, with average pump rates of 5m3/min in the 1$^{st}$ stage and 3m3/min in the second stage. Total proppant volumes averaged 150 MT in both stages.

Flowback data was recorded for durations up to 16 hours and long-term production data was collected over 180 days using surface measurements.

## Field Application Results and Discussion

Field trials in the Sultanate of Oman demonstrated the practical benefits of the neutral wettability proppant, validating laboratory predictions.

- **Overall Flowback Performance and Cleanup Time:** As flush of the frac fluid density making the well as overbalance, Nitrogen lifting is needed to kick-off the well which was done post milling the plug between 2 stages. Considering that part, we have seen the well response faster than other wells that was treated with standard HSP proppant. Also, the total frac fluids 400 m3 were recovered within 8 hours of flowback, and total water recovery reached over 792 m3 within 16 hours. A comparison with offset wells in the same area using native ceramic proppants showed a significant reduction in total cleanup time as below in Figure 10. The well with neutral wettability proppant cleaned up was approximately 23 hours faster (41% reduction) than the average cleanup time of 39 hours for the offset wells.
- **First Oil Breakthrough:** The faster fluid recovery directly translated to earlier first condensate breakthrough, and the first condensate was observed right after the plug milling within the 4.8 hours, with a sharp increase in oil recovery after 4 hours which significantly accelerated production compared to the wells treated using conventional/standard proppant type by 400% higher gas and condensate compared to the well expectations.

- **Production Data and Productivity Index:** Long term production data highlighted the benefits mentioned above. In a comparison of a neutral wet proppant well against a standard HSP proppant well. It showed a higher initial water cut that rapidly decreased, suggesting effective water block removal. Over a 180 days production period, the well produced approximately over 28% of cumulative condensate more than offset wells cumulative in the same period at higher reservoir pressure. Furthermore, the well maintained a higher flowing tubing pressure (THP) during its initial production phase, even with a smaller choke size (35%), indicating a lower drawdown pressure and thus a higher productivity index than the offset which could be explained with less restricted flow within the propped fracture.
- **Economic Impact:** The dramatic reduction in cleanup time led to significant economic savings. In one specific instance, saving approximately 23 hours of N2 lifting and SMS time translated to an approximate cost reduction, underscoring the substantial financial benefits of this technology.

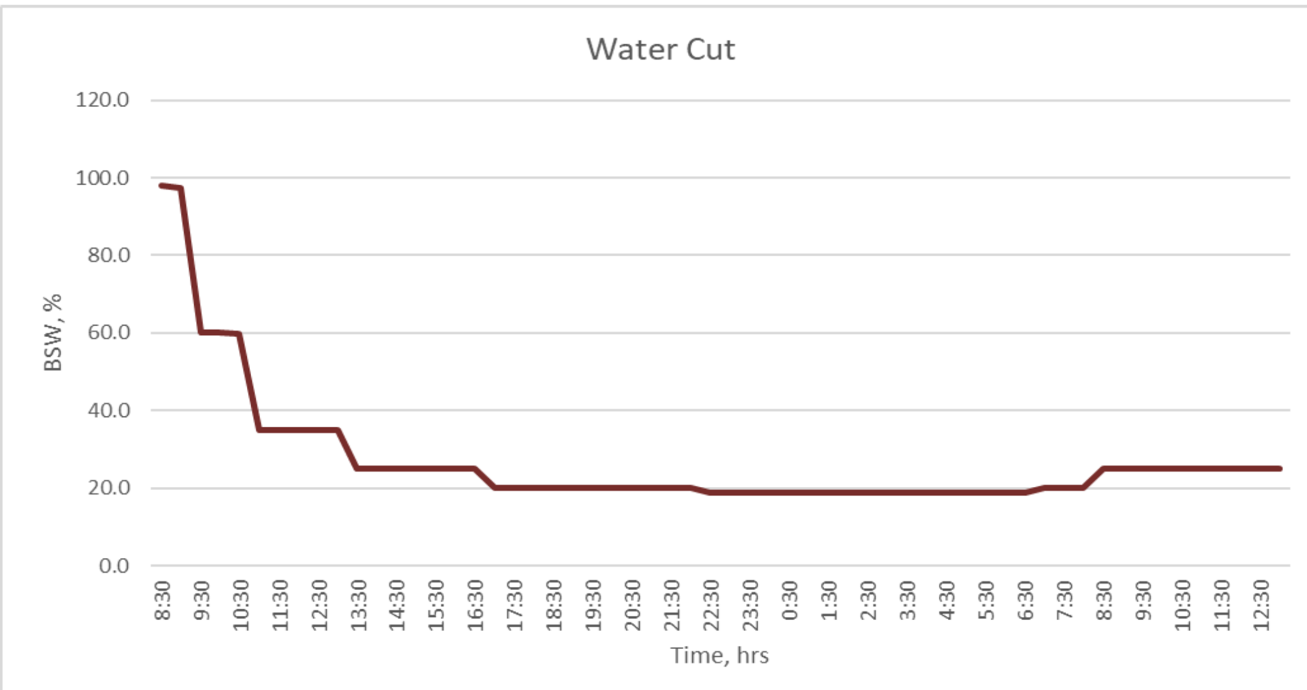


**Figure 9—Water Cut of the well was treated with Neutral Wet Proppant**

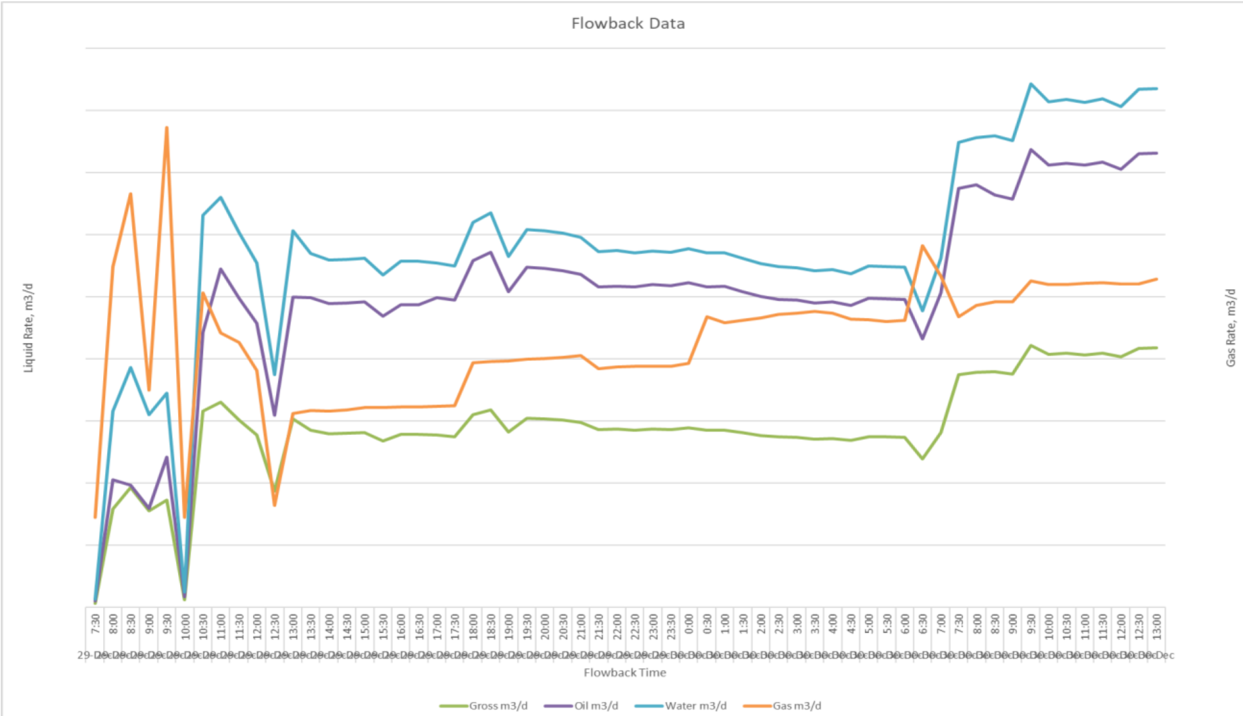


**Figure 10—Flow Back Data of the well was treated with Neutral Wet Proppant**

## Conclusion and Recommendation

This paper presents the implementation and successful field application of a novel neutral-wettability HSP proppant, developed to address the challenges of hydraulic fracturing in depleted wet gas wells, which are known for condensate banking challenges:

- The proppant is permanently surface modified with amphiphobic moieties, which minimize capillary pressure and enable efficient flow of both gas and condensate.
- Comprehensive laboratory testing validated the desired neutral wettability characteristics, improved proppant strength and properties, enhanced flow performance, with higher water and condensate recovery.
- The field case study demonstrated tangible operational benefits in significantly faster cleanup of fracturing fluids, earlier condensate breakthrough with a sustained improvements in well productivity which leading to substantial economic gains.
- This neutral wettability proppant marks a significant advancement in fracturing methodology for depleted wet gas wells by maximizing hydrocarbon recovery and enhancing the efficiency and profitability of well operations in challenging environments.

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## Glossary

Cf: Fracture Conductivity

Cp: Capillary Pressure (psi)
D50: Median Diameter (μm)
FTP: Flowing Tubing Pressure (psi)
ISO: International Organization for Standardization
kf: Fracture Permeability
LWC: Lightweight Ceramic
MD: Measured Depth
PPA: Pound Proppant Added
PPG: Pounds Per Gallon
PI: Productivity Index
Psi: Pounds Per Square Inch
SAMs: Self-Assembled Monolayers
γ: Interfacial Tension (dynes/cm)
θ: Contact Angle (degrees)
TVD: True Vertical Depth
ULWP: Ultra-Lightweight Proppant
wf: Fracture Width
FBI Fracture Barrier Index
HPHT High Pressure High Temperature

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